Semiconductor Classification

Device Numbering Systems

Semiconductor devices are classified by the manufacturer using a unique part numbering system. There are many systems in use, but here are some popular schemes in detail; the European based Pro-electron system, the American based JEDEC system and the Japenese based JIS system. Major manufacturers introduce their own schemes as well.

JEDEC Numbering System

The JEDEC system, (**J** oint **E**lectron **D**evice **E**ngineering **C**ouncil. This system has the following format:

digit, letter, serial number, [suffix]

Digit:
The first digit designates the amount of P-N junctions in the device. So a device starting with "2" would contain 2 P-N junctions and would most likely be either a transistor or a FET. Common part numbers are listed below:

1. Diodes
2. Bipolar transistors or Field Effect Transistors
3. Double Gate MOSFETS, SCR's
4. Opto Couplers

Letter:
The letter is always "N", and the remaining figures contain the device serial number.

Serial Number:
The serial number runs from 100 to 9999 and indicates nothing about the transistor.

Suffix:
If a suffix is present then this indicates the gain group as below:

A = low gain
B = medium gain
C = high gain
No suffix = ungrouped (any gain).

So for example, 1N4001 would be a diode and 3N201 would be a double gate MOSFET.

Pro-Electron Numbering System

This system uses the following format:

two letters, [letter], serial number, [suffix]

The 1st letter specifies the semiconductor material :

A Germanium
B Silicon
C Gallium Arsenide
R Compound Materials

The 2nd letter specifies the type of device :

A Diode, low power or signal
B Diode, variable capacitance
C Transistor, audio frequency low power
D Transistor, audio frequency power
E Diode, tunnel
F Transistor, high frequency low power
G Miscellaneous devices
H Diode, sensitive to magnetism
K Hall effect device
L Transistor, high frequency power
N Photocoupler
P Light detector
Q Light emitter
R Switching device, low power e.g. thyristor, diac, unijunction etc
S Transistor, low power switching
T Switching device power, e.g. thyristor, triac, etc.
U Transistor, switching power
W Surface acoustic wave device
X Diode, multiplier, e.g. varactor
Y Diode, rectifying
Z Diode, voltage reference

Third Letter:
If present this indicates that the device is intended for industrial or professional rather than commercial applications. It is usually a W,X,Y or Z. Examples- BFY51.

Serial Number:
The serial number runs from 100-9999.

Suffix:
If a suffix is present then this indicates the gain group as below:

A = low gain
B = medium gain
C = high gain
No suffix = ungrouped (any gain).

JIS System

The **J**apanese **I**ndustrial **S**tandard has the following format:

digit, two letters, serial number, [suffix]

Digit:
This indicates the amount of p-n junctions as in the JEDEC code.

Letters:
The letters indicate the intended application for the device according to the following code:

SA: PNP HF transistor	SB: PNP AF transistor
SC: NPN HF transistor	SD: NPN AF transistor
SE: Diodes	SF: Thyristors
SG: Gunn devices	SH: UJT
SJ: P-channel FET/MOSFET	SK: N-channel FET/MOSFET
SM: Triac	SQ: LED
SR: Rectifier	SS: Signal dioc
ST: Diodes	SV: Varicaps
SZ: Zener diodes	

Serial Number:
The serial number runs from 10-9999.

Suffix:
The (optional) suffix indicates that the type is approved for use by various Japanese organizations.

Major Manufacturers

Major manufacturers often produce their own code and numbering scheme for commercial reasons. The following abbreviations represent some of the larger semiconductor manufacturers:

MJ: Motorolla power, metal case
MJE: Motorolla power, plastic case
MPS: Motorolla low power, plastic case
MRF: Motorolla HF, VHF and microwave transistor
RCA: RCA
RCS: RCS
TIP: Texas Instruments power transistor (platic case)
TIPL: TI planar power transistor
TIS: TI small signal transistor (plastic case)
ZT: Ferranti
ZTX: Ferranti

Common examples include: TIP32A, MJE3055, ZTX302.

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